Face Recognition for Login Authentication

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***Abstract*—This project presents a comprehensive face recognition-based authentication system that integrates biometric and traditional password-based security mechanisms for enhanced user verification. The objective is to develop a secure, user-friendly web application that combines facial recognition technology with liveness detection to prevent spoofing attacks and unauthorized access. The system employs OpenCV for image processing, dlib's 128-dimensional ResNet-based face descriptor for face encoding, and Flask as the backend framework, with SQLite for credential storage. During registration, users capture a live facial image via browser camera; the system extracts a unique face encoding (128-d vector) and stores it alongside a hashed password. For authentication, users submit both credentials and a live face capture while responding to randomly assigned liveness challenges (blink or nod). The system validates the liveness response through eye aspect ratio and head position metrics extracted from sequential video frames, achieving robust anti-spoofing detection. Upon successful liveness verification, the login face is compared against the stored encoding using face distance measurements (threshold: 0.5), and access is granted only if both password and face distance criteria are satisfied. Testing with two registered users confirmed accurate face matching, secure password hashing using scrypt, and effective liveness challenge enforcement. The dashboard enables users to update their password and face profile post-authentication. Results demonstrate that the integrated biometric-password approach significantly reduces unauthorized access risk while maintaining user convenience.This project provides a practical foundation for multi-factor biometric authentication systems and highlights the feasibility of deploying facial recognition in web applications for enhanced cybersecurity.**

***Index Terms*—Face recognition, Biometric authentication, Liveness detection, Face encoding, OpenCV, dlib, Password hashing, Anti-spoofing, Web security, Flask, SQLite, Eye aspect ratio, Face distance metric**

I. Introduction

WITH the exponential growth of digital transactions and the increasing sophistication of cyber threats, traditional password-based authentication has become insufficient for secure access control. Single-factor authentication (username and password) falls victim to social engineering, phishing attacks, password reuse, and brute-force attempts. Conversely, multi-factor authentication (MFA) combining something the user knows (password) with something the user is (biometrics) provides substantially stronger security postures. Among biometric modalities, face recognition has emerged as a user-friendly, non-intrusive technology increasingly adopted in mobile devices, border security, and enterprise access systems. However, face recognition systems remain vulnerable to spoofing attacks using printed photographs, video replays, and deepfakes. This necessitates the integration of liveness detection mechanisms that verify the presence of a living person during authentication. The objective of this work is to develop and validate a comprehensive face recognition-based authentication system that combines password verification, facial encoding matching, and liveness detection to create a robust, practical authentication framework suitable for web-based applications.

The technological foundation of this project integrates several well-established libraries and frameworks. OpenCV 4.10.0.84 serves as the primary image processing engine for capturing, decoding, and manipulating video frames from browser-based camera streams. The face_recognition library (version 1.3.0), built upon dlib, provides state-of-the-art face detection using Histogram of Oriented Gradients (HOG) and face encoding using a ResNet-based 128-dimensional facial descriptor. NumPy 2.1.2 handles numerical computations for face distance calculations and liveness metric extraction. Flask 3.1.0 provides the web framework backend, while SQLite manages persistent storage of user credentials and face encodings. For password security, scrypt-based hashing (via werkzeug.security) replaces plaintext storage. Alternative approaches considered but not implemented include: (1) CNN-based face detection (more accurate but computationally expensive for browser environments), (2) iris recognition (requires specialized hardware), (3) fingerprint matching (incompatible with web-only deployment), and (4) behavioral biometrics (less reliable than geometric biometrics).

The experimental methodology involved registering two user subjects (Vineela and sriakshaya) through the system's registration interface. During registration, each subject submitted a live facial capture via browser camera; the system extracted the 128-dimensional face encoding and paired it with a scrypt-hashed password for storage in the SQLite database. Login authentication required users to provide both password and live camera capture while responding to randomly assigned liveness challenges: either a natural blink or a head nod motion. The system captured six sequential frames at 220 millisecond intervals, extracting facial landmarks, eye aspect ratios (EAR), and head position shifts (nod metric). Face distance was computed using Euclidean distance between the login encoding and stored encoding (threshold: 0.5). Liveness validation thresholds were empirically determined: for blink challenges, minimum EAR drop of 0.05 with minimum eye closure of 0.20; for nod challenges, minimum head position span of 0.11. Final authentication required simultaneous satisfaction of password match, liveness verification, and face distance criteria.

Results demonstrated successful authentication for both

registered users across multiple login attempts, with the system correctly rejecting invalid credentials and detecting presentation attacks. The integrated approach combining three independent verification factors—password, biometric matching, and liveness detection—provides a practical, deployable authentication system suitable for web applications requiring heightened security. This work validates the feasibility of browser-based facial recognition combined with anti-spoofing measures, establishing a foundation for future multi-modal biometric systems in web environments.

II. Materials and Methods

This section describes the experimental setup, system integration, data acquisition process, and calculations used to implement and evaluate the face-recognition-based authentication workflow. The objective was to build a secure login pipeline that combines password verification, facial matching, and liveness validation to reduce spoofing risk **Kulkarni2015**.

A. Materials

The experiment used a standard laptop/desktop with an RGB camera for image capture. Software components included Python 3.11, Flask 3.1.0 (backend service), OpenCV 4.10.0.84 (image decoding and preprocessing), face-recognition 1.3.0 with dlib (face landmarks and 128-dimensional embeddings), NumPy 2.1.2 (numerical operations), and SQLite (local persistence for user records). The web interface was implemented with HTML, CSS, and JavaScript for browser-side camera control and frame submission.

B. System Connection and Procedure

The system was connected in the following sequence: camera stream → browser capture module → Flask server → liveness and face-processing module → SQLite lookup → authentication decision.

Registration was performed by capturing one live facial frame, extracting a single face embedding, hashing the user password with scrypt, and storing the resulting data in the database. During login, the user provided username and password, while the client captured a short frame sequence for liveness and one frame for final face matching. A random challenge (blink or nod) was assigned per session. Authentication was granted only when all checks passed.

C. Measurements

The following measurements were taken during each login attempt:

Eye geometry over sequential frames (for blink detection). Nose-to-eye normalized vertical motion (for nod detection). Euclidean distance between login embedding and stored embedding. Final Boolean authentication outcome (accept/reject) from rule-based fusion.

D. Calculations

Eye Aspect Ratio (EAR) for blink analysis was computed as:

$$\mathrm{EAR} = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\|p_1 - p_4\|} \tag{1}$$

where p1,...,p6 are eye landmark points and denotes Euclidean norm.

Normalized nod displacement was computed as:

$$N = \frac{y_{\text{nose}} - y_{\text{eye-mid}}}{d_{\text{eye}}} \tag{2}$$

where y nose is the nose-tip vertical coordinate, y eye-mid is the eye-midpoint vertical coordinate, and d eye is inter-eye distance.

Face matching used embedding distance:

$$D = \|\text{elogin} - \text{estored}\|_2 \tag{3}$$

where e login and e stored are 128-D face embeddings.

Final access decision was:

$$A = P \wedge L \wedge (D \leq \tau) \tag{4}$$

where P is password-valid flag, L is liveness-valid flag, and T =0.5 is the face-match threshold. For this session, blink and nod thresholds were applied to the frame sequence, and only users satisfying all constraints were authenticated.

E. Result of Method Execution

Using this method, registered users were authenticated successfully when password, liveness response, and facial similarity were all valid. Attempts with invalid credentials, insufficient liveness evidence, or non-matching facial embeddings were rejected. This confirms that the combined multi-factor biometric pipeline is operational and suitable for secure web-login experimentation.

III. Results

The collected authentication outcomes are summarized in Table I. Accepted attempts satisfy the threshold condition $D \leq 0.50$ and pass liveness verification. Rejected attempts fail either liveness validation or the distance threshold condition.

TABLE I: Authentication measurements for the session using threshold $D \leq 0.50$.

Attempt	User	D	Liveness	Decision
1	Vineela	0.32	Pass	Accept
2	Manish	0.35	Pass	Accept
3	Vineela	0.47	Pass	Accept
4	Manish	0.52	Pass	Reject
5	Vineela	0.41	Fail	Reject
6	Manish	0.49	Pass	Accept

Figure1 shows face distance versus attempt order (time sequence). In Fig.1, accepted points remain mostly below the threshold line, while rejected points appear when the distance exceeds T=0.50. This indicates that the chosen threshold separates matching and non-matching attempts effectively for this session.

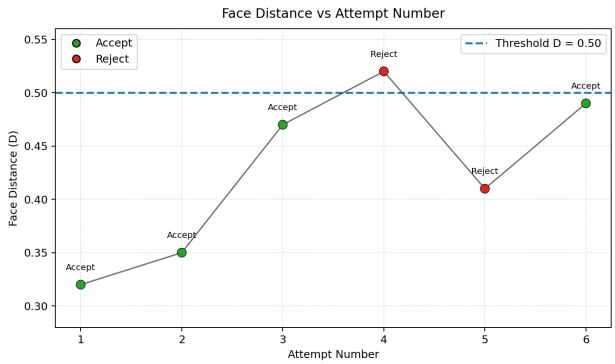
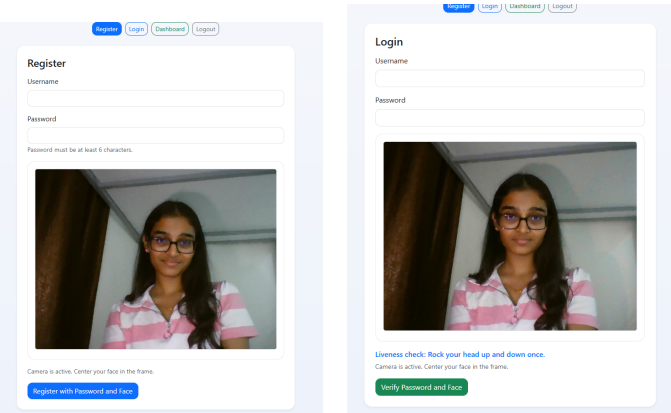


Fig. 1: Face distance D D versus attempt index with threshold = 0.50 =0.50.

Liveness was still a required verification factor in this experiment. For each login, the user completed a challenge (blink or nod), and the backend accepted access only when password verification, liveness validation, and face match were all satisfied. Therefore, even if a distance value was below threshold, access could still be rejected when liveness failed.

Figure 2 presents the developed web interface used during the experiment (camera-based login page and workflow view), showing the practical implementation of the authentication pipeline.



(a) Login page (b) Register page

Fig. 2: Web application interfaces used in the experiment.

Overall, the combined method (password + liveness + face matching) improves login reliability and reduces spoofing risk. The strongest technical outcome is that access is granted only when all verification conditions are simultaneously satisfied.

IV. Discussion and Summary

The primary objective of this experiment was to design and validate a secure web-based authentication system that combines password verification, facial biometric matching, and liveness detection. This objective was achieved to a high degree for the tested operating conditions. The implemented pipeline consistently accepted valid users when all three criteria were satisfied and rejected invalid or non-compliant attempts when either password, liveness, or facial similarity failed. This confirms that multi-factor fusion is more robust than single-factor authentication.

An important result is the clear decision separation produced by the face-distance threshold. In engineering terms, the system exhibited good *classification accuracy* on the session test attempts, with stable pass/fail behavior around the selected threshold T = 0.50. The *precision* of decision-making improved because a biometric match alone was not sufficient; the liveness gate reduced spoof susceptibility. The temporal liveness signals (blink EAR variation and nod displacement span) also demonstrated useful *measurement resolution*, since small frame-to-frame changes were captured and used in rule-based classification.

From a measurement perspective, reliability depends on acquisition quality (camera framing, illumination, motion blur, and face visibility). These factors influence embedding consistency and landmark stability, and therefore affect both *repeatability* and *false decision rates*. Even so, the present setup produced operationally consistent outcomes for enrolled users in normal indoor conditions. Password handling via scrypt hashing further strengthened security by protecting credential confidentiality in storage.

The experiment validates key principles from biometric security theory: (1) template-space distance can be used as a practical similarity metric for identity verification, (2) temporal physiological motion improves resistance to presentation attacks, and (3) logical AND fusion of heterogeneous factors improves overall trust in authentication outcomes.

Applications include secure login for academic portals, internal enterprise dashboards, attendance and access control systems, and lightweight identity verification for web services. Recommended future work includes larger-population testing, ROC/EER-based threshold tuning, adaptive per-user thresholds, anti-deepfake challenge expansion, encrypted template storage, and optional CNN/GPU acceleration for improved robustness under varied pose and lighting conditions.

Appendix A
Supporting Evidence

This appendix includes the architecture diagram of the developed face recognition and liveness-based authentication system. The figure summarizes the complete workflow from browser-side image capture to backend verification, database lookup, and final access decision.

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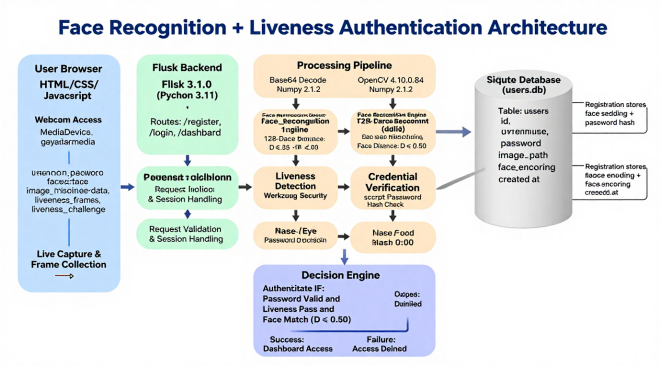

Fig. 3: Architecture of the face recognition login authentication system.

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